AEGIS / DEPLOYMENT NOTE 01 / 24 OCT 2026

Restormer, ready for the device test.

Frozen weights. Verified exports. Measured local evidence. Jetson results still pending.

DELIVERED	EVIDENCE
Restormer ONNX	Epoch 51 checkpoint; 26.13M parameters; static 1×3×256×448 input.
Frozen YOLOv8n ONNX	Official COCO weights; static 1×3×640×640 input; no detector finetuning.
Numerical checks	Restormer: four weather inputs, largest output error 3e-05. YOLO: three inputs, decoder agrees with Ultralytics.
Runtime & acceptance	FP16 engine builder, FP32 fallback, optional calibrated INT8, file/video/camera runner, checksums and timing logs.
Preserved	Training weights, restoration splits and Chitraksh’s contributed files are unchanged.

The deployment geometry is a deliberate tradeoff

Each image is resized with its aspect ratio preserved, padded to 256×448, restored, and returned to its original dimensions before detection. This reduces compute but may remove small-object details. The original 32.58 dB / 0.9297 native-resolution restoration score is not the score of this resized pipeline.

Local validation passed; edge acceptance has not happened

The laptop is CPU-only. No TensorRT engine, Jetson FPS, thermal limit, power draw or device memory result has been measured. Engines must be built and tested on the actual Jetson. The package does not contain invented hardware results.

Local CPU smoke timing: five ONNX model calls after two warmups, median 14.94 seconds. This short measurement checks execution only; it is not an edge-performance forecast.

02 / DOES RESTORATION HELP DETECTION?

Measure the downstream task.

72 real DAWN images: 24 fog, 24 rain, 24 snow. Images were selected by filename hash before scores were computed. All three views use the same frozen YOLOv8n and the same annotations.

Weather / view	mAP50 ↑	mAP50–95 ↑	Precision	Recall
all / original	44.85	29.97	58.57	40.07
all / resize only	44.52	29.25	54.45	**41.62**
all / restored	**49.09**	**32.86**	**72.02**	41.13
fog / original	40.91	26.99	47.36	**49.92**
fog / resize only	44.65	29.40	46.19	45.89
fog / restored	**56.30**	**37.11**	**89.54**	41.12
rain / original	**58.99**	**38.87**	63.87	**51.33**
rain / resize only	55.27	37.07	74.37	46.73
rain / restored	55.87	37.21	**76.35**	46.61
snow / original	47.72	**36.27**	38.97	**57.83**
snow / resize only	44.65	33.47	42.98	45.67
snow / restored	**47.92**	35.62	**47.96**	55.12

Across the pilot, restored minus original mAP50–95 is **+2.89 percentage points**; restored minus resize-only is **+3.60 points**. These are observed pilot differences, not a statistical superiority claim.

Bold marks the highest value within each weather group. Values are percentages. Precision/recall use the maximum-F1 confidence; mAP uses conf=.001, IoU=.7 NMS and up to 300 detections. Overall AP is pooled across the 72 images, not the arithmetic mean of weather AP.

03 / QUALITY CONTROLS & HANDOFF

A testable next step.

Data checks that make the comparison interpretable

All 704 available DAWN annotation pairs were checked: 300 fog, 200 rain and 204 snow images; 5,610 boxes across six road classes. Source TXT IDs were corrected to COCO IDs only in derived evaluation labels; every box was checked against VOC XML. One train object is outside the six-class scope. 17 exact duplicate groups were found; image counts must not be treated as independent scene counts.

No DAWN smoke detection labels are available in this project. COCO pretraining overlap is unverified. Pilot image choices, source hashes, class-level results and the full annotation audit are saved in JSON. The full 704-image evaluation is supplied as a separate resumable command; it was not run for this handoff.

Device acceptance sequence

STEP	PASS CONDITION
1 · Verify & preflight	Package checksums match. CUDA/TensorRT/OpenCV imports work. Actual hardware, free memory and software versions are recorded.
2 · Compile locally	Build one engine at a time. FP16 uses FP32-sensitive transformer math. No changes to system power modes or drivers.
3 · Precision gate	Finite output; restoration numerical agreement plus per-fixture PSNR drop ≤0.1 dB and SSIM drop ≤0.001. Reject failed candidates and retain FP32 fallback.
4 · Sustained run	Measure original, resize-only and restored paths with identical settings; record median/p95, wall throughput and tegrastats.
5 · Adoption decision	Check longer videos and untouched labelled data. Keep raw detection available. Small fixture tests alone do not establish field reliability.

Run from the extracted kit

bash scripts/deployment/edge_device.sh prepare
bash scripts/deployment/edge_device.sh benchmark

Read START_HERE.txt first for the JetPack system-packages environment, FP32 fallback, optional INT8 experiment and camera command. The archive has ONNX graphs, four-weather fixtures, 64 training-only calibration inputs, three DAWN samples and a SHA-256 manifest.

Sources: NVIDIA TensorRT engine compatibility and accuracy documentation; Ultralytics v8.3.0 source; DAWN Mendeley v3. Full links and third-party license notices are included in the handoff.

04 / FIXED EXAMPLES

Inspect the actual outputs.

One preselected pilot image per weather, shown before and after restoration. These are real degraded road images; no paired clean ground truth is available. A visually cleaner image is not proof of better object detection.

FOG / fog__mist-015

ORIGINAL



RESTORED



RAIN / rain__rain_storm-042

ORIGINAL



RESTORED



SNOW / snow__snow_storm-045

ORIGINAL



RESTORED

